Aditi Kanojia

akanojia@umich.edu | (623) 270-0556 | United States | linkedin.com/in/aditi-kanojia

Profile Summary

Cybersecurity Analyst with hands-on experience in detection engineering, cloud security, and network defense. Proven ability to operationalize threat intelligence, develop MITRE ATT&CK-aligned detection rules, and automate incident response workflows. Adept at securing cloud infrastructures (AWS, Azure, GCP), deploying SIEM/EDR platforms (Splunk, QRadar, CrowdStrike), and performing forensic log analysis. Passionate about improving security posture through scripting, threat hunting, and strategic collaboration across SOC and DevSecOps teams.

Education

M.S. in Cybersecurity and Information Assurance
University of Michigan, USA | Aug 2024 – Dec 2025

Post Graduate Certificate Program in Cybersecurity
Vellore Institute of Technology, Bangalore | Aug 2023 – Jun 2024

Bachelor of Computer Applications
Symbiosis International University, Pune | Jul 2019 – Jul 2022

Research Publications

"Water Quality Monitoring and Prediction using IoT and Analytics", IEEE (ICMACC 2024) DOI:10.1109/ICMACC62921.2024.10894063

- Designed a real-time, machine learning-enabled IoT framework to monitor environmental system failures, leveraging sensor fusion, anomaly detection, and time-series analysis to generate actionable alerts and improve water quality prediction in dynamic conditions.
- Integrated edge-device analytics with cloud-based telemetry pipelines to build a scalable, fault-tolerant architecture, enabling continuous environmental data ingestion, remote monitoring, and distributed model inference for resilient IoT deployments in resource-constrained environments.

Research Experience

Graduate Researcher, University of Michigan (Aug 2024 – Present)

- Simulated misconfigured Kubernetes RBAC policies to detect excessive access and lateral movement, improving container security by designing audit logic that exposes privilege escalation paths across tenant boundaries.
- Leveraged GCP Logging to monitor Pod access events and identify privilege misuse, generating structured JSON-based incident logs for threat modeling, forensics, and attack path mapping across namespaces.
- Built Python scripts to automate RBAC violation detection, enabling continuous least-privilege enforcement and early response to permission abuse in containerized, multi-tenant environments.
- Enhanced Kubernetes visibility by integrating policy violation alerts into existing SOC dashboards, streamlining detection and accelerating triage for access-related misconfigurations.

Fileless Malware Behavioral Analysis – *Independent Research Project, 2023*

- Reverse-engineered fileless malware to uncover stealthy techniques like reflective DLL injection and PowerShell abuse, bypassing traditional AV and EDR defenses using volatile memory forensics.
- Analyzed memory dumps to extract ephemeral artifacts and construct threat models of runtime behavior, enhancing incident triage workflows for advanced memory-resident malware.
- Authored custom YARA rules aligned with IOCs and ATT&CK tactics, integrating them into SOC pipelines to detect stealthy malware that evades disk-based scanning.
- Developed a detection strategy based on process hollowing and runtime injection behavior, improving SOC capabilities to identify evasive in-memory threats.

Ransomware Defense Strategy – *Independent Research Project, 2023*

- Executed ransomware simulations in sandboxed environments to analyze encryption onset, entropy behavior, and file system I/O patterns, creating behavioral profiles across ransomware variants to inform early-stage detection strategies and response workflows.
- Discovered consistent high-entropy spikes during encryption phases, using this insight to build custom rules for SIEM systems, enabling early threat identification before full data compromise occurred in test environments.
- Built an entropy-based anomaly detection engine in Python, integrating time-series analysis to flag sudden entropy surges and inform SOC teams of suspicious encryption activity.
- Connected detection engine outputs to SIEM alerting logic, enabling real-time notifications for ransomware behaviors and improving incident response timelines in simulated enterprise networks.

Cybercrime Data Analysis Using Tableau – *Independent Research Project, 2022*

- Cleaned and structured large-scale cybercrime datasets for threat intelligence analysis, categorizing attack vectors, frequency, geographic origin, and impacted sectors to support strategic planning and forensic investigations.
- Designed interactive Tableau dashboards to display real-time visualizations of cybercrime trends, enabling analysts to explore incident timelines, attack correlations, and anomaly clusters across regional and temporal dimensions.
- Applied statistical models to identify outlier patterns in cybercrime activity, integrating filters to allow drill-down exploration of high-risk indicators and aiding investigations into coordinated or persistent threat actors.
- Enabled SOC teams to correlate visualized threat data with known IOCs, improving threat response strategies and enhancing proactive detection through dynamic filtering and geographic analysis tools.

Research Lead, Affordable Agriculture Lab (Symbiosis International University) (Jul 2021 – Jul 2022)

- Led development of an IoT-based real-time water quality monitoring system with predictive analytics, improving sustainability for rural farming operations by generating actionable environmental insights.

- Built machine learning models for anomaly detection and predictive alerts, increasing reliability of early warnings for unsafe water quality and enabling corrective actions.
- Integrated edge-device analytics with a scalable cloud telemetry pipeline, creating a fault-tolerant system that supports continuous monitoring and remote access in low-resource environments.
- Published research in IEEE ICMACC 2024, presenting a secure and scalable architecture for water monitoring using IoT, analytics, and real-time distributed inference in constrained networks.

DeepCANvas: Neural System for Vehicle CAN Data Anomaly Detection (Published at IEEE IVAC)

- Engineered a hybrid CNN-Autoencoder model to detect anomalies in live CAN bus data, enhancing predictive diagnostics and anomaly-based detection for automotive cybersecurity.
- Developed standardized data preprocessing pipelines for real-time telemetry, ensuring compatibility across OEMs and reliable inference in noisy, edge environments with minimal resource usage.
- Used Vertex AI to deploy models with automated drift detection and online retraining, ensuring detection accuracy remains high as vehicle behavior patterns evolve in production systems.
- Published results in IEEE IVAC, contributing to automotive network security research by detecting intrusions and anomalies in CAN bus traffic using deep learning and real-time processing.

Technical Skills

Cloud Security: AWS (VPC, IAM, S3, CloudTrail), Azure Defender, GCP Networking, Terraform, Docker, Kubernetes
Detection & Response: Splunk, QRadar, CrowdStrike, Suricata, Snort, Sigma/YARA rules, MITRE ATT&CK
Threat Hunting & Automation: Python, Bash, PowerShell, Netmiko, Scapy, Ansible, Git
Network Security: VLANs, VPN (IPsec, OpenVPN), 802.1X, Firewall (Palo Alto, Cisco ASA), Wireshark, NetFlow
Compliance & Standards: NIST CSF, CIS Benchmarks, HIPAA, PCI-DSS, OWASP

Professional Experience

ML Data Associate | Amazon Aug 2022 – Jul 2023

- Analyzed AWS VPC Flow Logs and login metadata to detect insider threat behaviors, enhancing anomaly detection pipelines and improving detection precision by 30% through collaboration with security teams and refinement of model inputs.
- Integrated AWS CloudTrail and GuardDuty logs into Splunk, enriching event correlation and enhancing SOC visibility across multi-account cloud environments, contributing to faster incident response and improved threat contextualization during investigations.
- Collaborated with threat intelligence analysts to optimize detection signatures aligned with MITRE ATT&CK, significantly reducing false positives and enabling more accurate alert triage within the security operations center.
- Automated behavioral trend analysis using Python and Pandas, generating baselines from user activity logs to support proactive threat hunting and improve early warning capabilities across cloud-native workloads.

AWS Cloud Security Engineer | Zummit Infolabs Oct 2021 – Apr 2022

- Hardened AWS cloud infrastructure by auditing IAM policies, encrypting service communication, and segmenting networks with security groups to ensure least-privilege access and prevent lateral movement across sensitive workloads.
- Developed modular Terraform templates to provision CIS-compliant VPCs, EC2 instances, and S3 buckets, implementing standardized tagging and security baselines across staging and production cloud environments.
- Simulated red-team attack scenarios on EC2 workloads to test detection coverage, validate SOC alert rules, and refine cloud incident response plans based on findings.
- Integrated SAST and DAST security tools into CI/CD pipelines, enabling secure code validation and early vulnerability detection in cloud-native software development lifecycles.

Web Security Analyst | Meredevelop Jun 2021 – Dec 2021

- Performed penetration testing using OWASP ZAP and Burp Suite to identify critical web application vulnerabilities including XSS, CSRF, and IDOR, supporting secure coding practices and risk-based vulnerability management.
- Created detailed threat models and collaborated with engineering teams to deploy WAF rules and mitigate discovered security issues, improving web application security posture and reducing public exposure.
- Authored internal secure coding standards and led developer workshops, driving adoption of secure development practices and reducing repeat vulnerabilities in production codebases.
- Configured and deployed HTTPS and SSL/TLS certificates across web assets to ensure encrypted data transmission, mitigating risk from MITM attacks and meeting compliance requirements.

Certifications

- CompTIA Security+ (In Progress)
- Certified Cybersecurity (CC) – ISC2 (Udemy Verified)
- Certified Information Security & Ethical Hacking – CRISIL

Leadership and Contributions

- **Chair, IEEE Women in Engineering (WIE) Affinity Group (SICSR)** – Led an IEEE student affinity group (Symbiosis Institute, SBA11434) to promote technical leadership, mentorship, and inclusion for women in cybersecurity and engineering.
- **STEM Education Volunteer, IEEE SAMPARK Program** – Taught foundational computer skills to underprivileged children in rural communities, advancing digital literacy and early STEM education through weekly workshops. (Membership:)
- **Diversity in Cybersecurity Initiatives** – Organized hands-on workshops, peer mentoring sessions, and lab training programs to increase diversity in cybersecurity research, empowering more women to pursue technical careers.